# Modular Operator Discovery TSP Report

## Overview

- Condition count: 7.
- Best final transfer gap: phase7_full_solver_evolution.
- Best held-out TSPLIB gap: phase7_full_solver_evolution.
- Surviving modular candidates: none.

## Run Metadata

- run_name: run_20260520_082243_s
- started_at_local: 2026-05-20 08:22:43
- finished_at_local: 2026-05-20 08:53:48
- duration_hhmm: 00:31
- duration_seconds: 1865.176
- seed_offset: 18000
- replicate_label: s
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

## Condition Comparison

| Condition | Mode | Replay | Selection | Final TSPLIB Gap | Final Family Gap | Final Transfer Gap | Mean Novelty | Mean Complexity | Surviving Candidate | Transplant Delta | Pareto Runtime Inflation |
|---|---|---|---|---|---|---|---|---|---|---|---|
| phase7_baseline_heuristic_only | baseline_only | none | score_only | 0.235058 | 0.054968 | 0.121317 | 0.0 | 0.0 | False | 0.0 | 0.0 |
| phase7_full_solver_evolution | full_solver | none | score_only | 0.081984 | 0.0 | 0.052172 | 0.821136 | 0.76 | False | 0.0 | 0.0 |
| phase7_modular_operator_evolution | modular_operator | none | score_only | 0.235667 | 0.054968 | 0.121542 | 0.792608 | 0.445 | False | 0.001735 | 0.344792 |
| phase7_modular_operator_random_replay | modular_operator | random | score_only | 0.235058 | 0.054968 | 0.121317 | 0.8657 | 0.44 | False | 0.003056 | 0.007338 |
| phase7_modular_operator_diversity_residual_replay | modular_operator | diversity_residual | score_only | 0.235871 | 0.054968 | 0.121617 | 0.83759 | 0.44 | False | 0.00543 | -0.035494 |

| Condition | Mode | Replay | Selection | Final TSPLIB Gap | Final Family Gap | Final Transfer Gap | Mean Novelty | Mean Complexity | Surviving Candidate | Transplant Delta | Pareto Runtime Inflation |
|---|---|---|---|---|---|---|---|---|---|---|---|
| phase7_modular_operator_compression_pressure | modular_operator | none | novelty_gate | 0.235058 | 0.050827 | 0.118702 | 0.864706 | 0.445 | False | 0.005437 | 2.055607 |
| phase7_modular_operator_pareto_selection | modular_operator | none | pareto | 0.235058 | 0.054968 | 0.121317 | 0.866546 | 0.4425 | False | 0.003072 | -0.005965 |

## Condition Notes

### phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

### phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.081984, family holdout 0.0, combined 0.052172.
- Accepted novelty 0.821136, complexity 0.76, and adaptation efficiency 0.047651.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

### phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.792608, complexity 0.445, and adaptation efficiency 0.00491.
- Last-epoch runtime 139.384025 ms and distance evaluations 9649.75.
- Validation: surviving False, transplant delta 0.001735, positive scaffolds 1, Pareto runtime inflation 0.344792.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.234827, "gap_delta": 0.000224, "runtime_inflation": 0.344792, "same_gap_faster": false}

### phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.8657, complexity 0.44, and adaptation efficiency 0.006874.
- Last-epoch runtime 38.8161 ms and distance evaluations 3810.75.

- Validation: surviving False, transplant delta 0.003056, positive scaffolds 0, Pareto runtime inflation 0.007338.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": 0.007338, "same_gap_faster": false}

### phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235871, family holdout 0.054968, combined 0.121617.
- Accepted novelty 0.83759, complexity 0.44, and adaptation efficiency 0.004617.
- Last-epoch runtime 42.092525 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.00543, positive scaffolds 1, Pareto runtime inflation -0.035494.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.006899, "gap_delta": 0.000299, "runtime_inflation": -0.035494, "same_gap_faster": true}

### phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.050827, combined 0.118702.
- Accepted novelty 0.864706, complexity 0.445, and adaptation efficiency 0.019173.
- Last-epoch runtime 42.7546 ms and distance evaluations 4333.0.
- Validation: surviving False, transplant delta 0.005437, positive scaffolds 2, Pareto runtime inflation 2.055607.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 1.878498, "gap_delta": -0.003363, "runtime_inflation": 2.055607, "same_gap_faster": false}

### phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.866546, complexity 0.4425, and adaptation efficiency 0.019023.
- Last-epoch runtime 39.878675 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.003072, positive scaffolds 0, Pareto runtime inflation -0.005965.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": -0.005965, "same_gap_faster": true}

## Judge Appendix

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# TSP Modular-Operator Discovery Suite Review

## Summary of Main Results

| Condition | Final Transfer Gap | Final Family Gap | Final TSPLIB Gap | Surviving Candidate | Transplant Mean Gap Delta | Ablation Support | Notes |
|---|---|---|---|---|---|---|---|
| phase7_baseline_heuristic_only | 0.121317 | 0.054968 | 0.235058 | No | 0.0 | N/A | Baseline only, no operator |
| phase7_full_solver_evolution | 0.052172 | 0.0 | 0.081984 | No | 0.0 | No | Full solver evolution, no modular operators |
| phase7_modular_operator_evolution | 0.121542 | 0.054968 | 0.235667 | No | 0.001735 | No | Candidate pruner; no clear family or heldout gain; disabled variant no worse; runtime inflation high (0.34) |
| phase7_modular_operator_random_replay | 0.121317 | 0.054968 | 0.235058 | No | 0.003056 | No | Perturbation operator; no family/heldout gap improvement; ablation shows no operator effect |
| phase7_modular_operator_diversity_residual_replay | 0.121617 | 0.054968 | 0.235871 | No | 0.00543 | No | Candidate pruner variant; no family/heldout improvement; disabled variant no worse |
| phase7_modular_operator_compression_pressure | 0.118702 | 0.050827 | 0.235058 | No | 0.005437 | No | Restart controller; slight family gain on two_cluster_bottleneck; runtime inflation very high (2.05); ablation disabled hurts (gap +0.0033) |
| phase7_modular_operator_pareto_selection | 0.121317 | 0.054968 | 0.235058 | No | 0.003072 | No | Perturbation variant; no family/heldout gap improvement; ablation disabled no effect |

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## Conservative Interpretation

### Transfer Evidence (Held-out TSPLIB and Family Gaps)

- None of the modular operators demonstrated improvements in held-out TSPLIB or family gaps relative to the host baseline or baseline heuristic.
- Slight positive or negative gap deltas on heldout TSPLIB and families are near zero, indicating no meaningful transfer gains.
- Only the compression pressure condition showed a modest family gain (-0.024425 gap delta) on the "two_cluster_bottleneck_tsp" family, but this came with high runtime inflation and no heldout TSPLIB gap improvement.

### Transplant Checks

- Transplant mean gap deltas for modular operators are small positive numbers (~0.0017 to 0.0054), indicating no consistent or clear improvement under transplant.
- Some scaffolds show minor positive delta, but no scaffold or transplant condition demonstrates robust gap improvements.
- Operators generally fail to transplant cleanly to some scaffolds (notably clustered_local_search), limiting claims of structural reusability.

### Ablation Checks

- Ablation results show that disabling operators does not worsen solution gaps.
- Best variant gap deltas vs. full operator are zero or slightly negative, indicating no ablation-supported operator effect.
- No operator shows convincing ablation signal confirming the operator’s contribution.

### Pareto & Runtime Tradeoffs

- Runtime inflation for modular operators is often high, e.g., ~0.34 for candidate pruners and 2.05 for restart controller.
- No Pareto improvements observed: better gap at same runtime or same gap at faster runtime.
- Reduced runtime variants exist but without gap improvements.

### Operator Survival & Discovery

- No operator survived modular transplant, ablation, and family transfer checks simultaneously.
- No operator classified as "surviving_candidate."
- No clear operator discovery supported by scaffold and family tests.

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## Summary of Individual Operator Findings

1. **Candidate Pruner (phase7_modular_operator_evolution & diversity residual replay)**

- Core idea: Adaptive candidate-list pruning adapting neighborhood size with structural cues.
- Validation: No family or heldout TSPLIB gap improvement; disabling does not worsen gap; runtime overhead noticeable.
- Transplant: Minor positive gap deltas but no survival.
- Conclusion: Not a reusable or impactful operator under modular testing framework.

2. **Perturbation Operators (random replay & pareto selection)**

- Core idea: Deterministic structured perturbations (double-bridge, segment reversal) for stagnation escape.
- Validation: No family or heldout TSPLIB gap improvement; no ablation effect; no transplant survival.
- Runtime inflation negligible or negative but no benefit.
- Conclusion: Not validated as reusable stagnation escape operators.

3. **Restart Controller (compression pressure)**

- Core idea: Adaptive restart count to escape bottlenecks, specifically two-cluster.
- Validation: Small family gap gain on two_cluster_bottleneck_tsp, but no TSPLIB gain; runtime inflation significant.
- Ablation disables operator leads to worse gap (~+0.0033), suggesting some modest operator effect.
- Transplant shows minor positive gains on 2 scaffolds.
- Conclusion: Marginal candidate with some family signal but poor transplant and high runtime cost; not surviving candidate.

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## Final Conclusion

- None of the tested modular operators qualify as surviving candidates due to lack of consistent transfer gap improvements on held-out TSPLIB and family tests and failure of ablation and transplant validation.
- No operator can be claimed as reusable or structurally novel in the TSP context based on this data.
- No modular-operator "discovery" is supported.
- Full-solver performance gains in the best full_solver condition do not correspond to reusable modular operator effects.
- The suite correctly finds some candidate ideas (adaptive pruning, structured perturbation, adaptive restart), but these do not survive conservative validation to merit claims of effective modular operator discovery or transfer.

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**Recommendation:**

Focus on improving operator validation pipelines to ensure ablation effects are detectable and transplant checks are more robust before claiming reusable operator structures or algorithm discovery from modular-operator evolution results.